



R.M.K. ENGINEERING COLLEGE
(An Autonomous Institution)
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Department Of Information Technology

A PROJECT REPORT

**“WHATNEXT VISION MOTORS: SHAPING THE FUTURE OF MOBILITY
WITH INNOVATION AND EXCELLENCE”**

Submitted By

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B.Tech IT-B

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

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R.M.K. ENGINEERING COLLEGE

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1.ABSTRACT

WhatsNext Vision Motors is an automotive organization that aims to improve customer experience and operational efficiency through Salesforce CRM technology. The primary objective of this project is to develop a centralized vehicle management and customer relationship system that simplifies vehicle sales, dealer management, test-drive scheduling and service request handling. The Salesforce application stores and manages important business information including vehicle details, vehicle stock, pricing, dealer information, customer details, orders, test drives and service requests. By maintaining these details within a centralized CRM platform, the organization can reduce manual data handling and improve accessibility of information. One of the important features of the project is automated dealer assignment. Based on the customer's location, the system is designed to identify the appropriate dealer and associate the order with the dealer. This improves the convenience of customers and reduces manual effort for employees. The system also addresses vehicle stock availability. Orders should not be processed for vehicles that are unavailable. The automation and Apex components are designed to validate stock availability and maintain accurate order status. The project also includes automated test-drive reminders. Customers receive reminders before their scheduled test drive, improving communication and reducing missed appointments. Batch and scheduled processing can be used to periodically evaluate stock and order information. This provides a scalable approach for handling large volumes of records. Overall, the WhatsNext Vision Motors Salesforce CRM solution provides a centralized, automated and scalable platform for managing automotive business operations.

2.OBJECTIVES

The main objective of the WhatsNext Vision Motors Salesforce CRM project is to develop a centralized CRM solution that improves vehicle sales operations and customer relationship management.

Major Objectives

1. Centralized Vehicle Management

- Store vehicle information in Salesforce.
- Maintain model, price, stock and availability information.
- Provide easy access to vehicle records.

2. Dealer Management

- Maintain authorized dealer information.
- Store dealer location and contact details.
- Associate vehicles and customer orders with appropriate dealers.

3. Customer Management

- Maintain customer information in a centralized system.
- Store contact information and address.
- Track customer interactions related to vehicles.

4. Order Management

- Track vehicle orders.
- Maintain order status.
- Prevent invalid orders when stock is unavailable.

5. Test Drive Management

- Schedule test drives.
- Associate customers with vehicles.
- Send automated reminders.

6. Service Management

- Track vehicle service requests.
- Maintain service dates and issue descriptions.
- Monitor service request status.

7. Process Automation

- Reduce manual work.
- Automate repetitive business processes.
- Improve data accuracy.

8. Reporting

- Provide useful business reports.
- Support management decision-making.

3. PROBLEM STATEMENT

Automotive organizations manage large volumes of information involving customers, vehicles, dealers, inventory, orders and service activities. When these activities are handled manually or across multiple disconnected systems, several operational problems can occur.

Identified Problems

1. Difficulty in maintaining centralized information

Vehicle, customer and dealer information may be stored separately, making it difficult for employees to access complete information quickly.

2. Stock availability issues

Customers may attempt to order vehicles that are unavailable. This can create fulfillment problems and negatively affect customer satisfaction.

3. Manual dealer assignment

Assigning customers to dealers manually can consume employee time and may result in

inefficient dealer allocation.

4. Missed test-drive reminders

Without automation, customers may not receive timely reminders about scheduled test drives.

5. Manual order-status updates

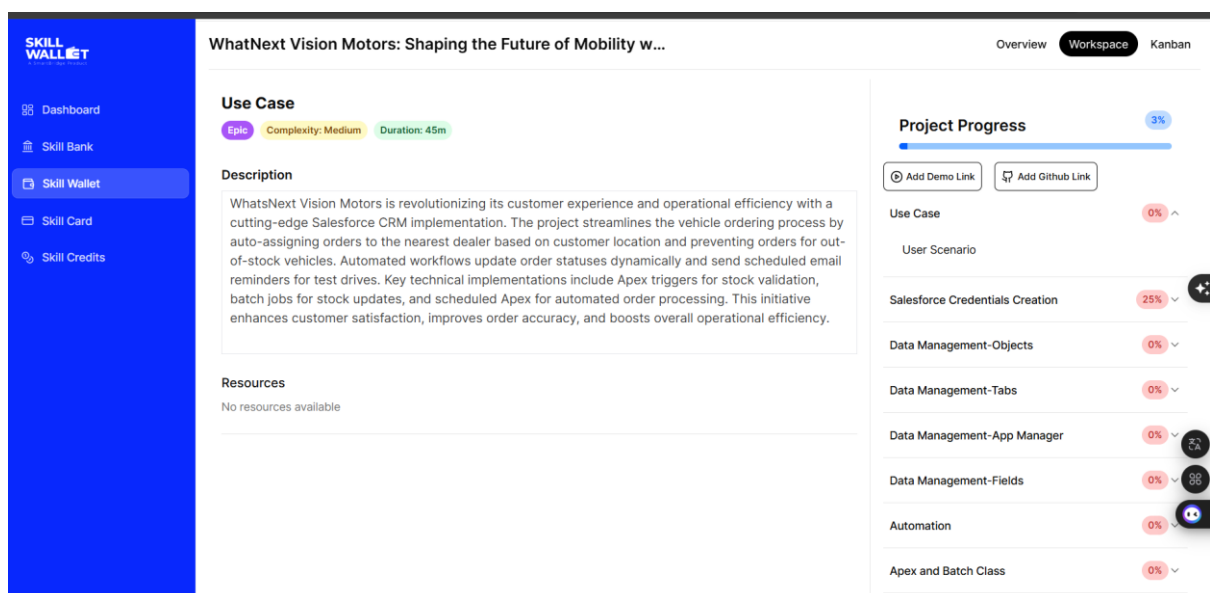
Employees may need to repeatedly check vehicle availability and update order statuses manually.

6. Limited reporting visibility

Management may find it difficult to obtain real-time insights into inventory, sales orders, customer activity and service requests.

Problem Statement

The organization therefore requires a centralized CRM platform capable of managing automotive data while automatically enforcing business rules and executing repetitive operational tasks.



4. PROPOSED SYSTEM

The proposed solution uses Salesforce as the central CRM platform for WhatsNext Vision Motors.

The system combines declarative Salesforce configuration with programmatic development to create a scalable business application.

Proposed Solution

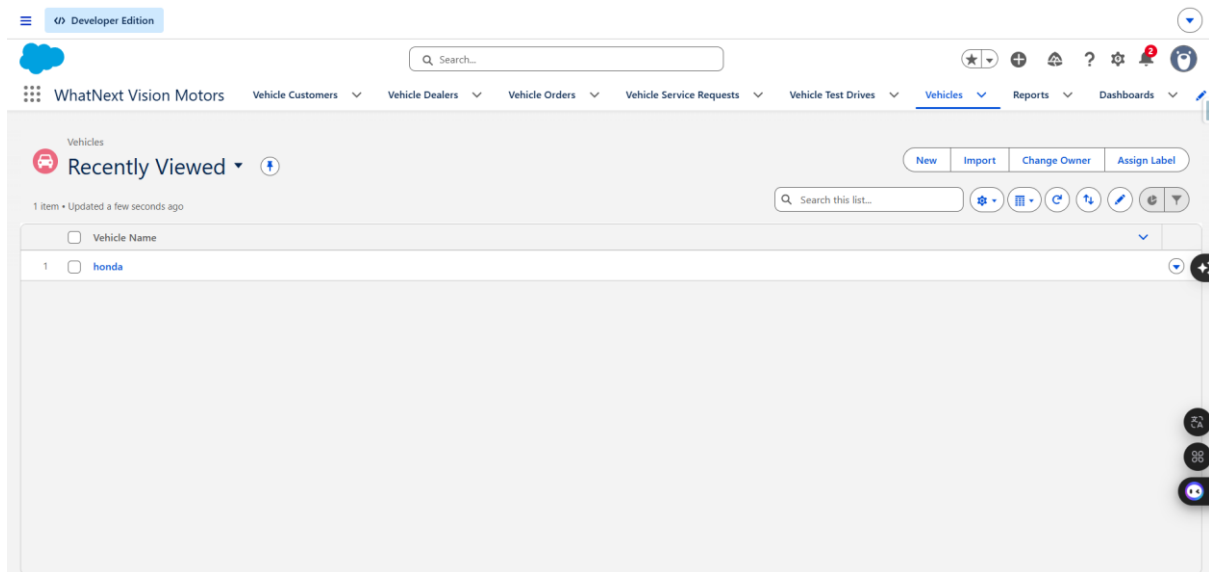
The Salesforce solution consists of:

- Custom automotive objects

- Custom fields
- Lookup relationships
- Lightning application
- Record-triggered flows
- Scheduled paths
- Email automation
- Apex triggers
- Trigger handler classes
- Batch Apex
- Scheduled Apex
- Reports
- Dashboards
- Security configuration
- Testing framework

Project Scope

Area	Scope
Vehicle Management	Vehicle details, pricing and stock
Dealer Management	Dealer details and locations
Customer Management	Customer contact and address information
Order Management	Vehicle orders and status
Test Drives	Scheduling and reminders
Service	Service requests and status
Automation	Flows and scheduled processing
Development	Apex and triggers
Bulk Processing	Batch Apex
Analytics	Reports and dashboards
Security	Profiles, permissions and sharing



5. REQUIREMENTS ANALYSIS

Requirements analysis establishes the functionality and technical capabilities required to implement the CRM system.

Functional Requirements

Vehicle Management

The system shall:

- Store vehicle name and model.
- Maintain vehicle pricing.
- Track stock quantity.
- Maintain availability status.
- Associate vehicles with dealers.

Dealer Management

The system shall:

- Store dealer name.
- Maintain dealer location.
- Store dealer contact information.
- Provide dealer identification through dealer code.

Customer Management

The system shall:

- Store customer name.

- Store email and phone.
- Maintain customer address.
- Store preferred vehicle type.

Order Management

The system shall:

- Associate an order with a customer.
- Associate an order with a vehicle.
- Maintain order date.
- Track order status.
- Prevent or appropriately handle unavailable vehicle orders.

Test Drive Management

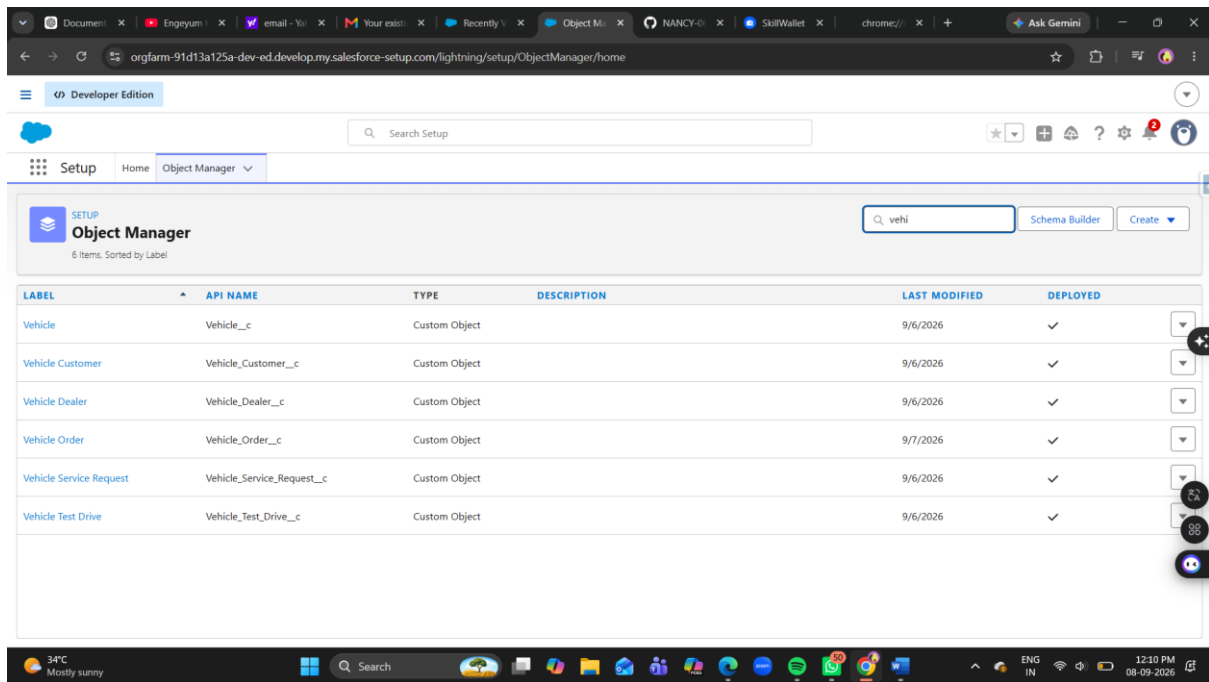
The system shall:

- Store customer and vehicle information.
- Maintain scheduled test-drive date.
- Track test-drive status.
- Send reminder communication.

Service Management

The system shall:

- Record customer service requests.
- Associate requests with vehicles.
- Record issue descriptions.
- Track service status.



6. SYSTEM ARCHITECTURE

The WhatsNext Vision Motors solution follows a layered Salesforce architecture.

Architecture Layers

User Interface Layer

- Lightning Experience
- Lightning App
- Record Pages
- Reports
- Dashboards



Automation Layer

- Record-Triggered Flow
- Scheduled Paths
- Email Actions
- Validation Rules



Application Logic Layer

- Apex Classes
- Apex Triggers
- Trigger Handlers
- Batch Apex
- Scheduled Apex

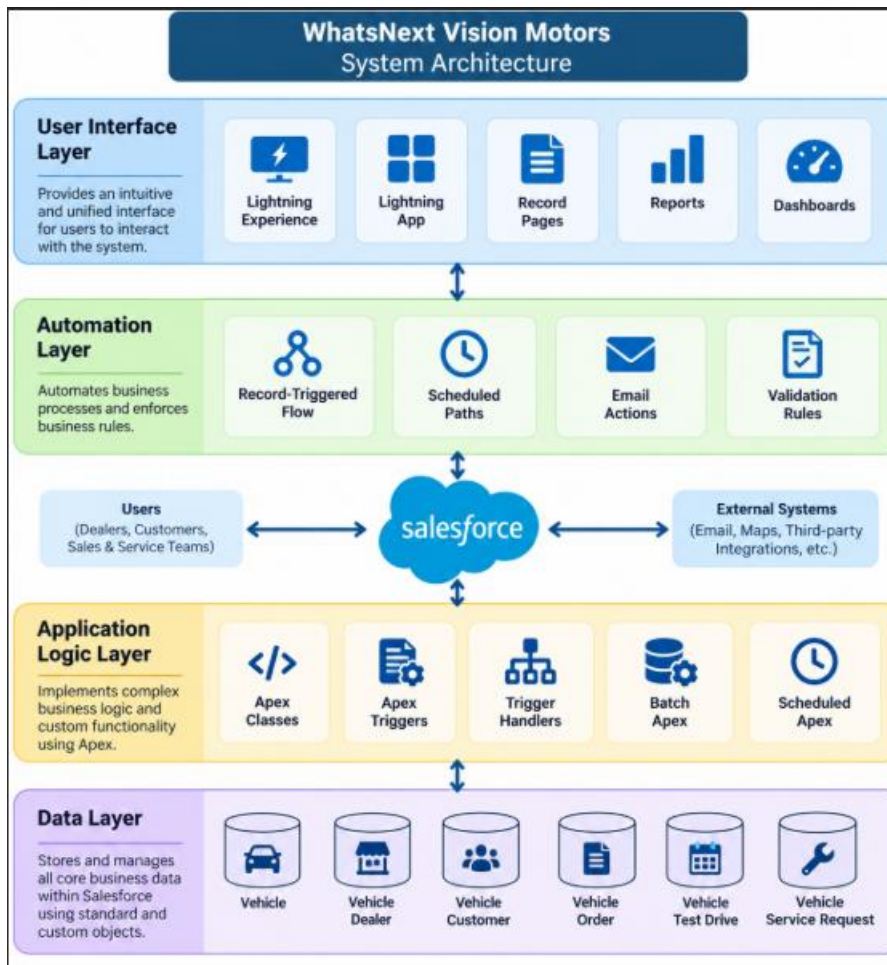


Data Layer

- Vehicle
- Vehicle Dealer
- Vehicle Customer
- Vehicle Order
- Vehicle Test Drive
- Vehicle Service Request

Architecture Objective

The architecture separates user interaction, automation, business logic and data management. This improves maintainability and allows future features to be added without significantly redesigning the existing solution



7. DATA MODEL

The data model defines how information is organized and connected within Salesforce.

The project uses custom objects representing the major entities involved in the automotive business process.

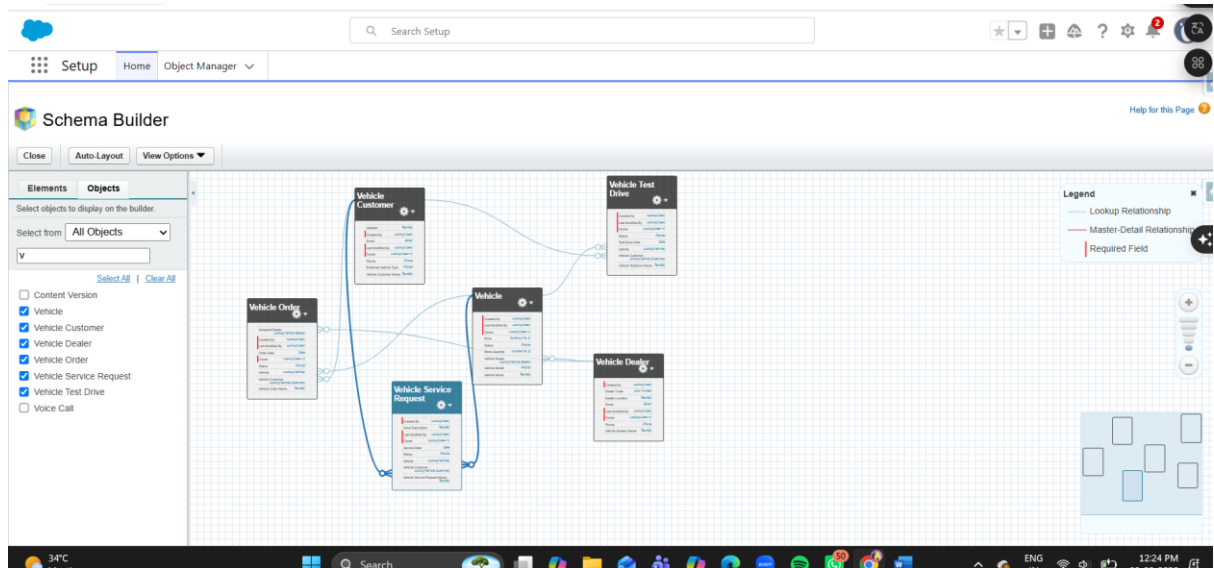
Core Objects

1. Vehicle
2. Vehicle Dealer
3. Vehicle Customer
4. Vehicle Order
5. Vehicle Test Drive
6. Vehicle Service Request

Relationship Overview

- A customer can have multiple vehicle orders.
- A customer can schedule multiple test drives.

- A customer can create multiple service requests.
- A vehicle can be associated with dealer information.
- A vehicle can appear in multiple order and test-drive records.
- Orders connect customers with selected vehicles.



8. SALESFORCE CUSTOM OBJECTS

Vehicle

The Vehicle object represents the automotive inventory maintained by WhatsNext Vision Motors.

Important information includes vehicle name, model, stock quantity, price, dealer and availability status.

Vehicle Dealer

The Vehicle Dealer object maintains information about authorized dealers, including location and contact details.

Vehicle Customer

The Vehicle Customer object stores customer information used throughout the CRM process.

Vehicle Order

The Vehicle Order object represents a customer's request to purchase a particular vehicle.

Vehicle Test Drive

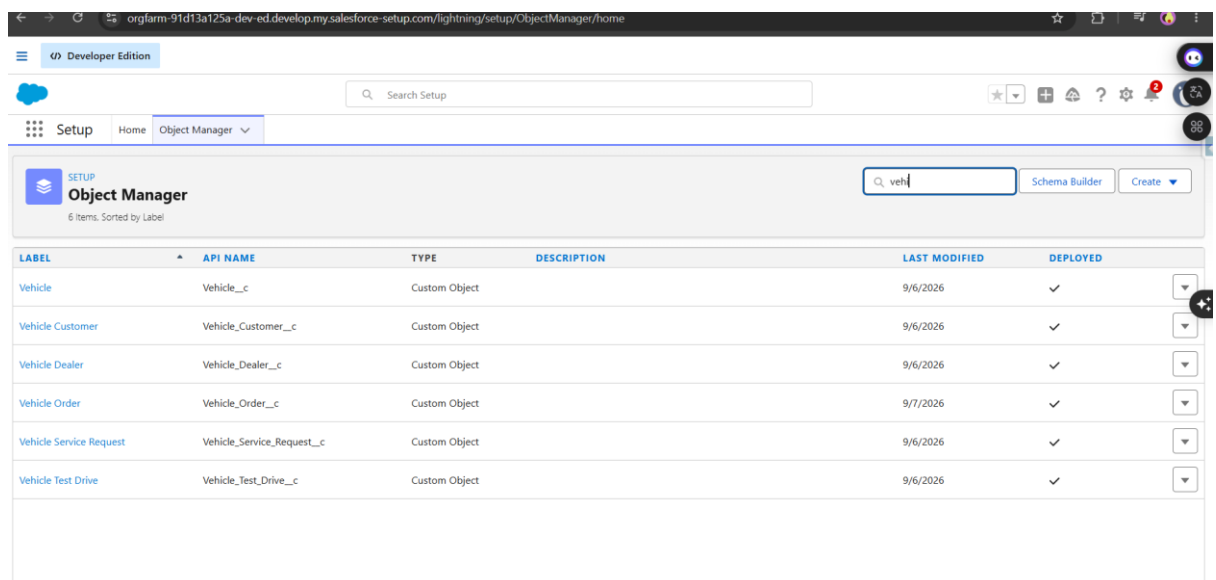
The Vehicle Test Drive object records scheduled test-drive activities.

Vehicle Service Request

The Vehicle Service Request object captures customer service requirements related to a vehicle.

Object Summary

Object	Purpose
Vehicle	Vehicle and inventory management
Vehicle Dealer	Dealer management
Vehicle Customer	Customer management
Vehicle Order	Vehicle sales orders
Vehicle Test Drive	Test-drive scheduling
Vehicle Service Request	Service management



9. FIELDS AND RELATIONSHIPS

The Salesforce data model uses custom fields to represent business information.

Vehicle Fields

Field	Data Type
Vehicle Name	Text
Vehicle Model	Picklist
Stock Quantity	Number
Price	Currency

Field	Data Type
-------	-----------

Dealer	Lookup
--------	--------

Status	Picklist
--------	----------

Dealer Fields

Field	Data Type
-------	-----------

Dealer Name	Text
-------------	------

Dealer Location	Text
-----------------	------

Dealer Code	Auto Number
-------------	-------------

Phone	Phone
-------	-------

Email	Email
-------	-------

Customer Fields

Field	Data Type
-------	-----------

Customer Name	Text
---------------	------

Email	Email
-------	-------

Phone	Phone
-------	-------

Address	Text
---------	------

Preferred Vehicle Type	Picklist
------------------------	----------

Order Fields

Field	Data Type
-------	-----------

Customer	Lookup
----------	--------

Vehicle	Lookup
---------	--------

Order Date	Date
------------	------

Status	Picklist
--------	----------

Relationship Design

Lookup relationships are used where records need to reference related business entities while maintaining flexibility in the data model.

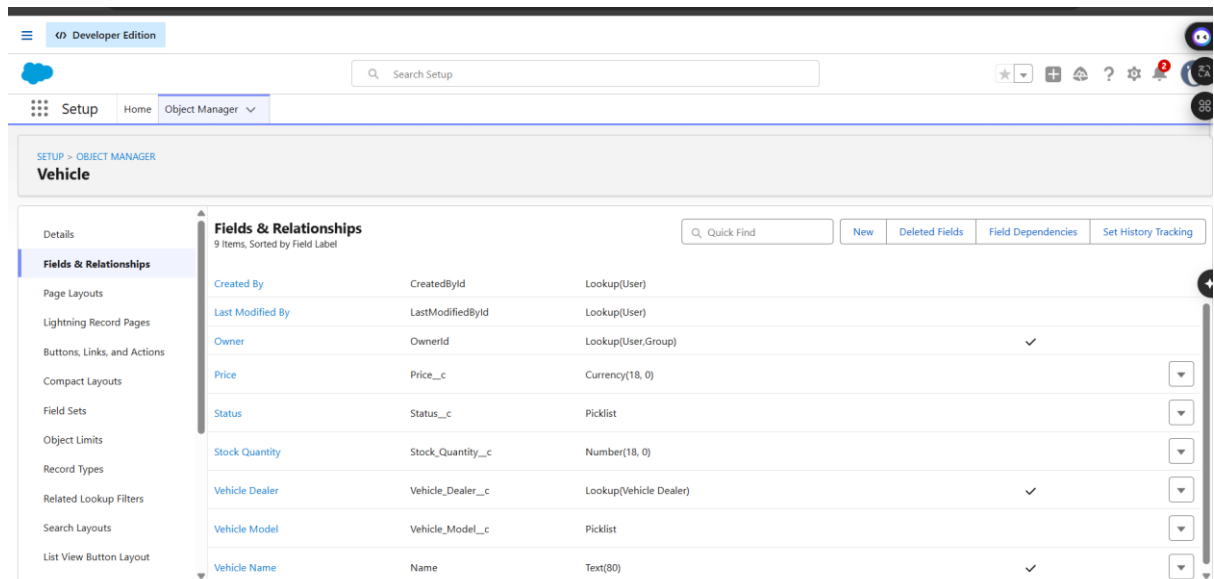


Figure 14.1 – Vehicle Fields and Relationships

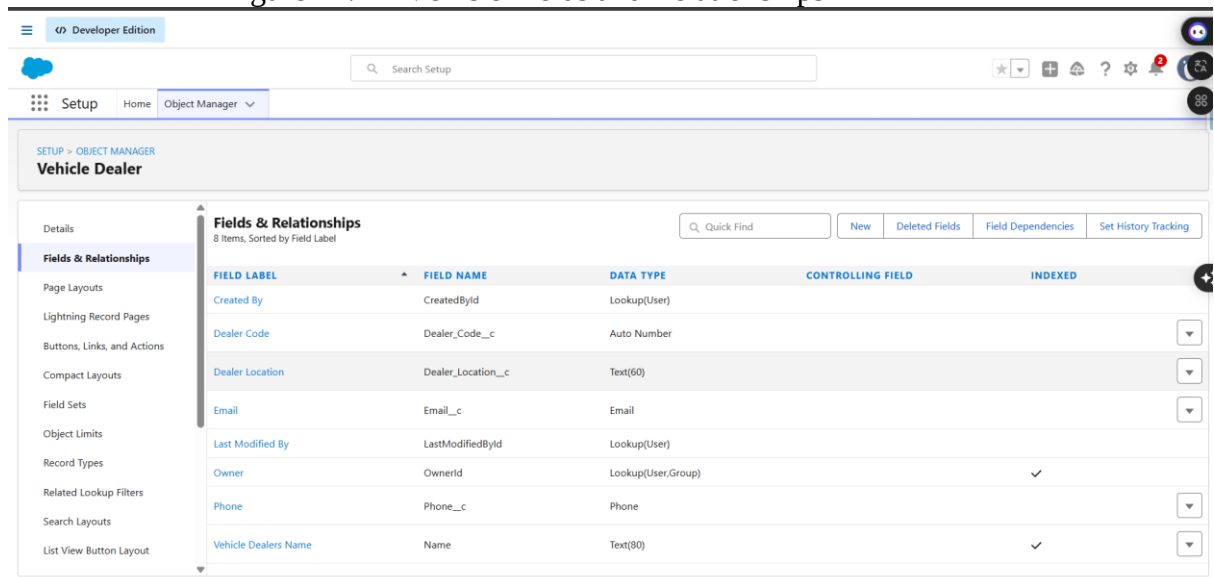


Figure 14.2 – Dealer Lookup Relationship

SALESFORCE ENVIRONMENT

The application is developed within a Salesforce Developer environment.

Environment Requirements

- Windows 10 or later / macOS / Linux
- 64-bit operating system
- Minimum 8 GB RAM
- Recommended 16 GB RAM
- Minimum 10 GB available storage
- Stable internet connection
- Salesforce Developer Org

Lightning Application

A dedicated Lightning application named:

WhatsNext Vision Motors

is configured to provide centralized access to the CRM functionality.

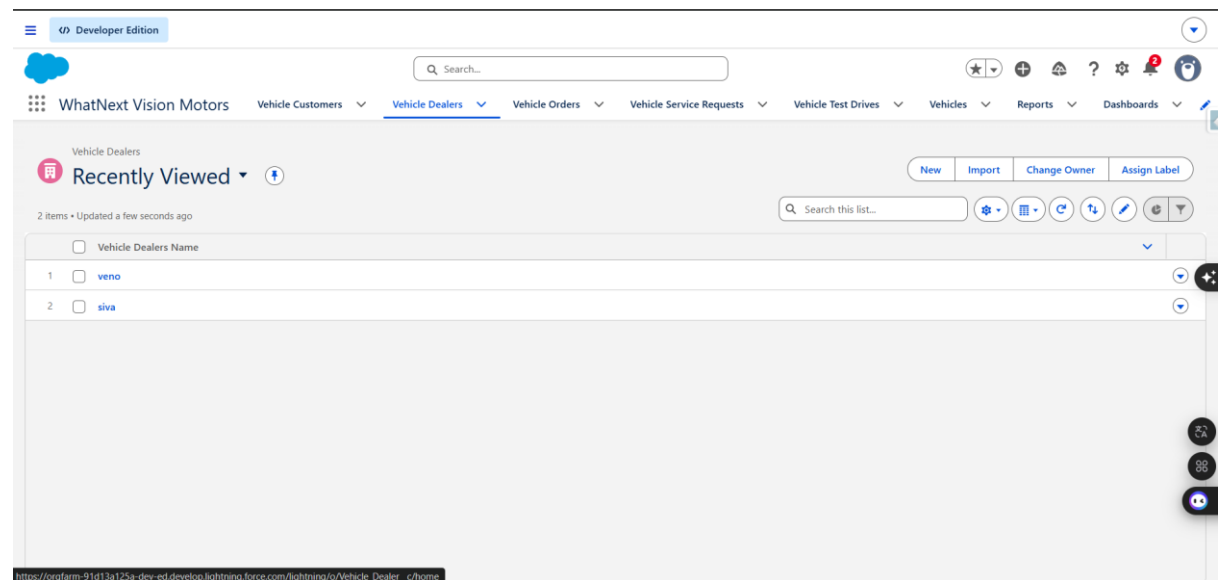
Navigation Items

The application can contain:

- Vehicles
- Vehicle Dealers
- Vehicle Customers
- Vehicle Orders
- Test Drives
- Service Requests
- Reports
- Dashboards

Application Objective

The Lightning application provides users with a consistent navigation experience and brings related business processes together within a single workspace.



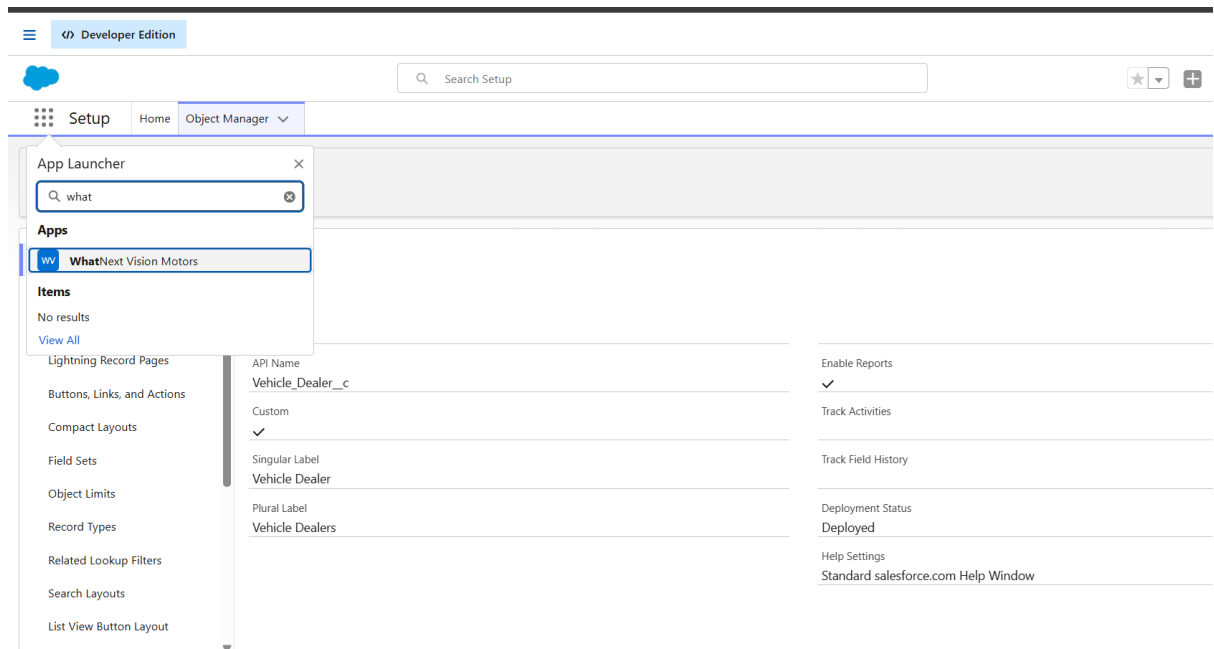


Figure 15.2 – WhatsNext Vision Motors Lightning App

PHASE 1 – REQUIREMENT ANALYSIS & PLANNING

The first phase focused on understanding the automotive business requirements and translating them into Salesforce functionality.

Planning Activities

- Identify business entities.
- Define CRM users and responsibilities.
- Determine required custom objects.
- Identify field requirements.
- Define relationships between objects.
- Identify automation opportunities.
- Define security requirements.
- Plan reporting requirements.
- Define testing strategy.

Security Model

Security is considered at multiple Salesforce levels:

Object-Level Security

Determines which objects users can access.

Field-Level Security

Controls whether specific fields can be viewed or edited.

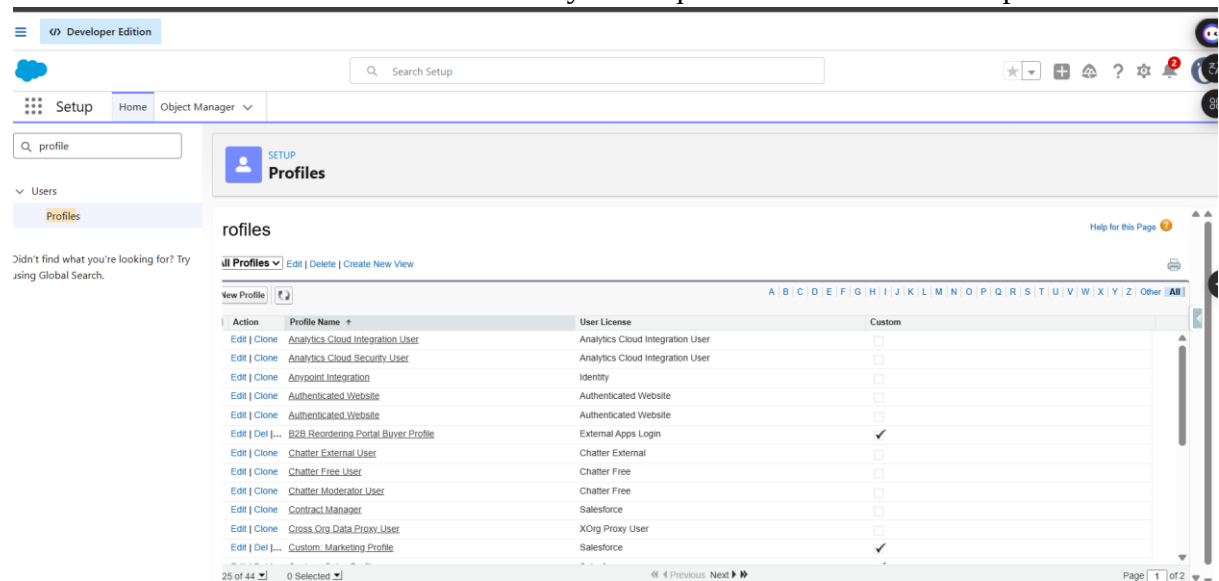
Record-Level Security

Controls access to individual records using sharing mechanisms and role hierarchy.

Permission Sets

Additional permissions can be assigned to users without modifying their base profile.

Security Objective The security design aims to ensure that users have appropriate access to the data and functionality required for their responsibilities.



PHASE 2 – SALESFORCE DEVELOPMENT: BACKEND & CONFIGURATION

The backend configuration establishes the foundation required for the application.

Major Components

- Custom objects
- Custom fields
- Picklists
- Lookup relationships
- Validation rules
- Page layouts
- Record-triggered flows
- Apex triggers
- Apex classes

- Batch Apex
- Scheduled Apex

Validation Strategy

Validation rules are used to prevent invalid data from being saved when a business condition is violated.

For example, stock quantity should not contain an invalid negative value.

Example Business Rule

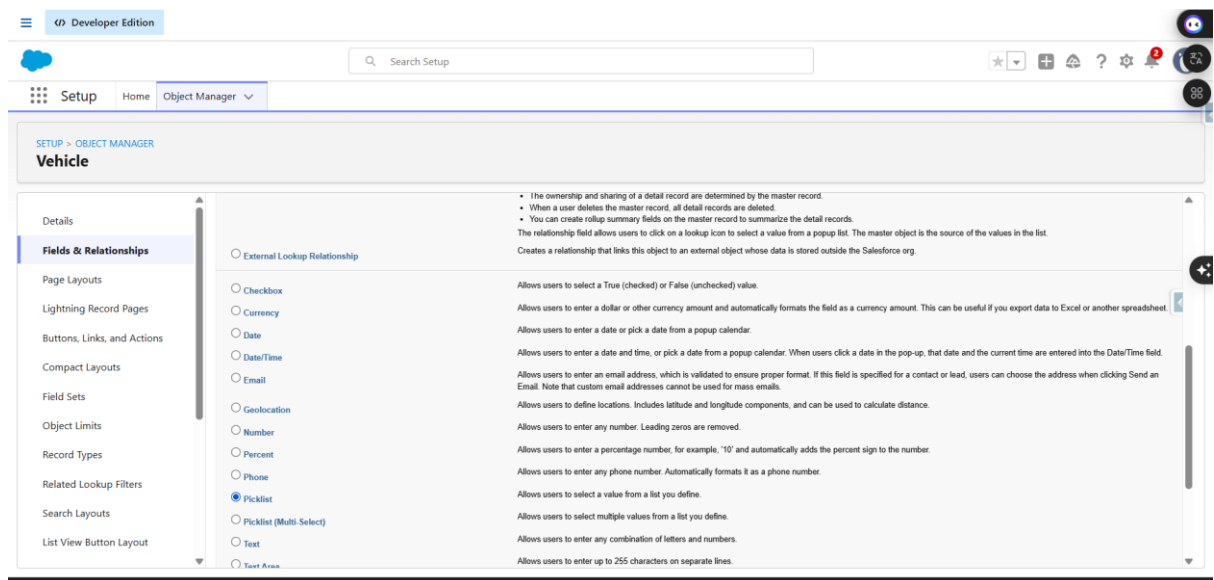
Stock Quantity ≥ 0

This ensures that inventory data remains logically consistent.

Configuration Principles

The project follows these principles:

- Use declarative automation where appropriate.
- Use Apex when complex logic is required.
- Avoid unnecessary automation duplication.
- Keep business logic modular.
- Design for bulk processing.
- Maintain clear naming conventions.



13. PROCESS AUTOMATION

Automation is a major component of the WhatsNext Vision Motors CRM.

Salesforce Flow Builder is used to automate repetitive business processes and reduce manual

intervention.

Automation Requirements

The system supports automation for:

1. Dealer assignment
2. Test-drive reminders
3. Order processing
4. Stock-related operations

Record-Triggered Flow

A record-triggered flow executes automatically when a record is created or updated and meets the configured entry conditions.

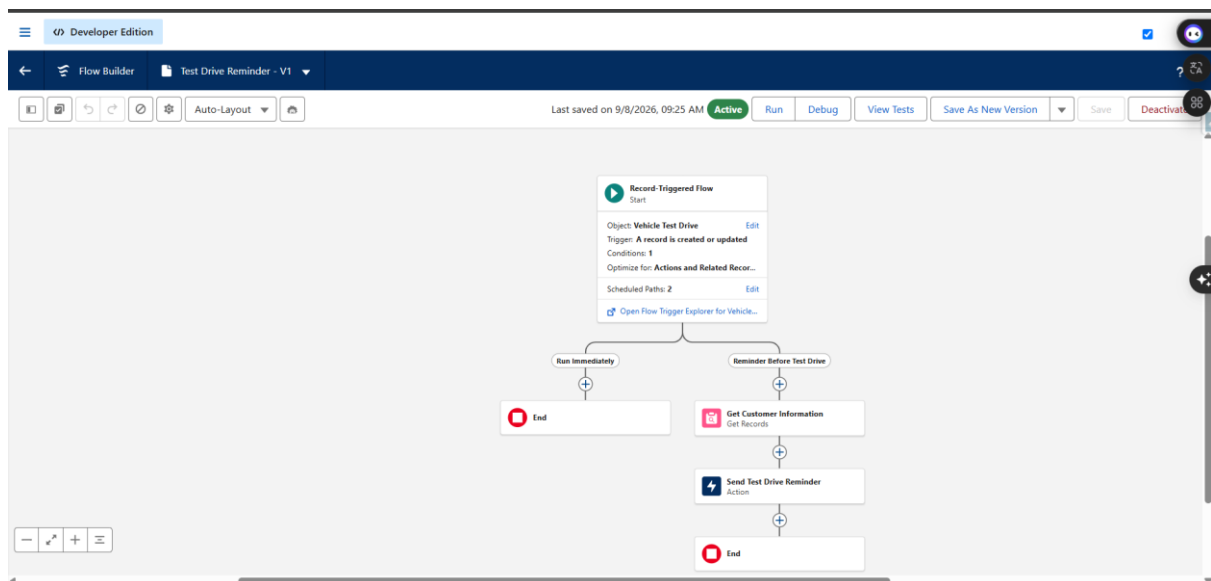
Benefits

- Reduces manual processing.
- Improves consistency.
- Provides faster response.
- Supports business rules.
- Reduces repetitive administrative work.

Flow Design

The automation follows a structured pattern:

Record Event → Retrieve Required Data → Evaluate Business Logic → Perform Action



14. AUTOMATIC DEALER ASSIGNMENT

One of the key requirements is to automatically associate an appropriate dealer with the customer order.

The automation uses customer location information to identify the corresponding dealer.

Business Logic

When a vehicle order is created with a pending status:

1. The system identifies the associated customer.
2. Customer address information is retrieved.
3. Dealer information is searched based on location.
4. The relevant dealer record is identified.
5. The order/dealer relationship is updated according to the configured design.

Flow Components

- Record-Triggered Flow
- Get Customer Information
- Get Nearest Dealer
- Update Records

Business Benefit

Automatic dealer assignment reduces the need for employees to manually search for dealers and improves the speed of order processing.

15. AUTOMATED TEST DRIVE REMINDER

Customer communication is improved through an automated test-drive reminder process.

When a test-drive record is scheduled, the flow creates a scheduled path based on the test-drive date.

Process

Test Drive Created/Updated

↓

Status = Scheduled

↓

Scheduled Path: One Day Before Test Drive

↓

Retrieve Customer Email



Send Reminder Email

Reminder Content

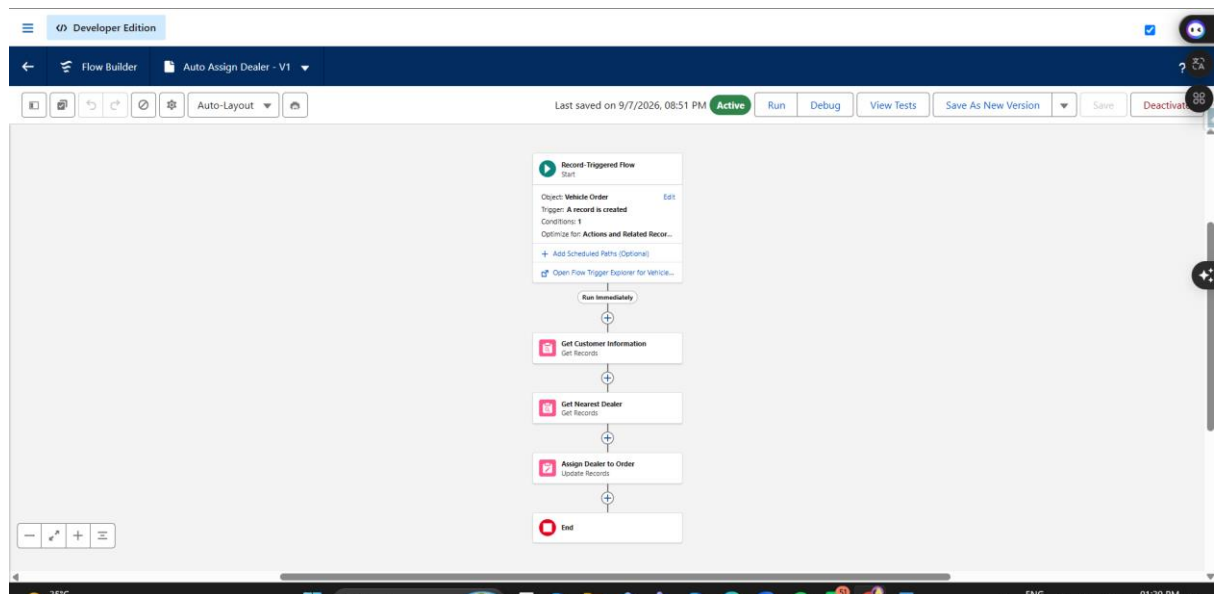
Subject:

Reminder: Your Test Drive is Tomorrow!

The email is automatically sent to the customer's registered email address.

Business Benefits

- Reduces missed appointments.
- Improves customer communication.
- Eliminates manual reminder activities.
- Provides a consistent customer experience.



16. APEX AND APEX TRIGGERS

Apex is used where business logic requires programmatic processing beyond standard declarative configuration.

The project uses Apex triggers to enforce important automotive business rules.

Trigger Responsibilities

The trigger logic can support:

- Vehicle stock validation.
- Order processing.
- Dealer assignment where programmatic logic is required.

- Business-rule enforcement.

Stock Validation

Before an order is accepted for processing, the system checks vehicle availability.

Conceptually:

Vehicle Stock > 0 → Order can proceed

Vehicle Stock = 0 → Order requires appropriate handling

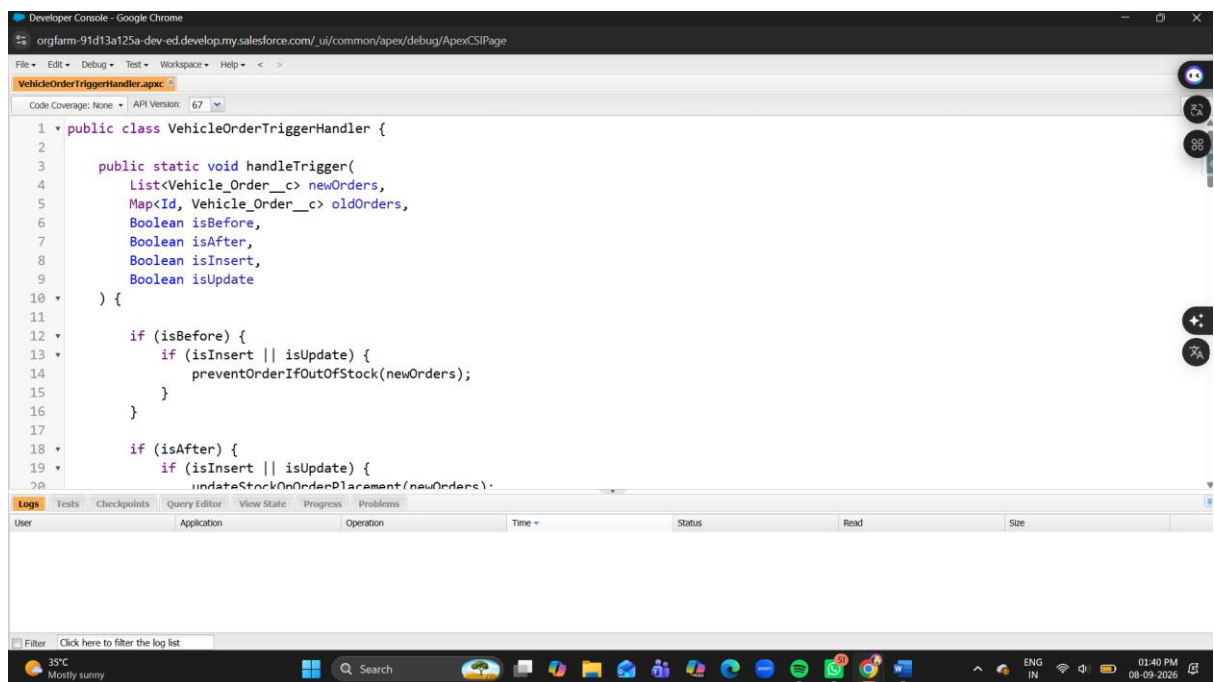
Trigger Handler Pattern

A trigger handler approach is recommended to keep the trigger itself lightweight.

Trigger → Handler Class → Business Logic

This approach provides:

- Better maintainability.
- Improved code organization.
- Easier testing.
- Reduced complexity.
- Reusable business logic.



17. BATCH APEX

Batch Apex is designed to process large numbers of Salesforce records asynchronously.

In the WhatsNext Vision Motors project, Batch Apex can be used for periodic vehicle stock evaluation and inventory-related processing.

Batch Processing Lifecycle

Start



Query Records



Execute Batch Scope



Update Records



Finish

Stock Processing

The batch process can evaluate vehicle stock levels and update availability information according to configured business rules.

For example:

Stock Quantity Availability

Greater than 0 Available

Equal to 0 Out of Stock

Advantages

- Handles large data volumes.
- Reduces limitations associated with synchronous processing.
- Supports scheduled execution.
- Improves bulk data processing capability.

Notifications

Where configured, scheduled processing can also support notifications related to stock replenishment or order processing.

```

1 global class VehicleOrderBatch
2 implements Database.Batchable<SObject> {
3
4     global Database.QueryLocator start(
5         Database.BatchableContext bc
6     ) {
7
8         return Database.getQueryLocator([
9             SELECT Id, Status__c, Vehicle__c
10            FROM Vehicle_Order__c
11            WHERE Status__c = 'Pending'
12            AND Vehicle__c != null
13        ]);
14     }
15
16     global void execute(
17         Database.BatchableContext bc,
18         List<Vehicle_Order__c> orderList
19     ) {
20

```

18. SCHEDULED APEX

Scheduled Apex allows Apex code to execute automatically at a predefined time.

The project uses Scheduled Apex as part of the automated processing architecture for recurring business operations.

Purpose

Scheduled execution can be used for:

- Periodic order processing.
- Stock evaluation.
- Inventory status synchronization.
- Scheduled notifications.

Processing Model

Schedule



Scheduled Apex



Batch / Business Logic



Update

Salesforce

Records



```
1 global class VehicleOrderBatchScheduler
2     implements Schedulable {
3
4     global void execute(SchedulableContext sc) {
5
6         VehicleOrderBatch batchJob =
7             new VehicleOrderBatch();
8
9         Database.executeBatch(batchJob, 50);
10    }
11 }
```

Business Benefits

Scheduled automation reduces the dependency on manual administrative activities and ensures that recurring processes are executed consistently.

Difference Between Batch and Scheduled Apex

Batch Apex

Processes large data volumes

Uses batch execution

Suitable for bulk processing

Can be invoked by Scheduled Apex

Scheduled Apex

Executes at scheduled time

Uses predefined schedule

Suitable for recurring jobs

Can invoke batch processing

PHASE 3 – UI/UX DEVELOPMENT & CUSTOMIZATION

The user interface is designed using Salesforce Lightning Experience.

The goal is to provide users with a simple and consistent interface for managing automotive operations.

UI Components

- Lightning App
- Lightning Record Pages
- Page Layouts
- Related Lists
- Tabs

- Navigation Items
- Reports
- Dashboards

Page Layout Design

Page layouts organize fields and related information into logical sections.

User Experience Principles

- Keep frequently used fields visible.
- Group related information.
- Use meaningful labels.
- Reduce unnecessary fields.
- Provide relevant related lists.
- Maintain consistent navigation.

Lightning Pages

Lightning App Builder can be used to customize record and application pages according to user requirements.

19. REPORTS AND DASHBOARDS

Salesforce reporting provides management with visibility into CRM activities.

Recommended Reports

1. Vehicle Inventory Report
2. Available Vehicles Report
3. Out-of-Stock Vehicles Report
4. Vehicle Orders by Status
5. Orders by Dealer
6. Scheduled Test Drives
7. Completed Test Drives
8. Service Requests by Status

Dashboard KPIs

A management dashboard can display:

- Total Vehicles
- Available Vehicles
- Out-of-Stock Vehicles

- Total Orders
- Pending Orders
- Confirmed Orders
- Scheduled Test Drives
- Open Service Requests

Business Value

Reports and dashboards convert operational CRM data into meaningful information that can support management decisions.

PHASE 4 – DATA MIGRATION, TESTING & SECURITY

Data Migration

Sample or existing business data can be loaded into Salesforce using appropriate Salesforce data-loading tools.

Data Loading Sequence

A recommended sequence is:

1. Dealer data
2. Vehicle data
3. Customer data
4. Vehicle orders
5. Test drives
6. Service requests

This sequence helps maintain lookup relationships because referenced records should exist before dependent records are loaded.

Data Quality

Data quality should be verified for:

- Duplicate records
- Missing mandatory fields
- Invalid email addresses
- Invalid phone numbers
- Incorrect stock values
- Incorrect lookup relationships

Security

Security controls include:

- Profiles
- Permission Sets
- Role Hierarchy
- Sharing Rules
- Field-Level Security
- Object permissions

20. TESTING STRATEGY

Testing ensures that Salesforce configuration and development components perform according to the defined requirements.

Testing was considered at multiple levels.

Functional Testing

Verifies that individual features behave correctly.

Integration Testing

Verifies that related Salesforce components work together.

Automation Testing

Validates Flow, Apex and scheduled processes.

Negative Testing

Attempts invalid operations to verify that business rules prevent inappropriate transactions.

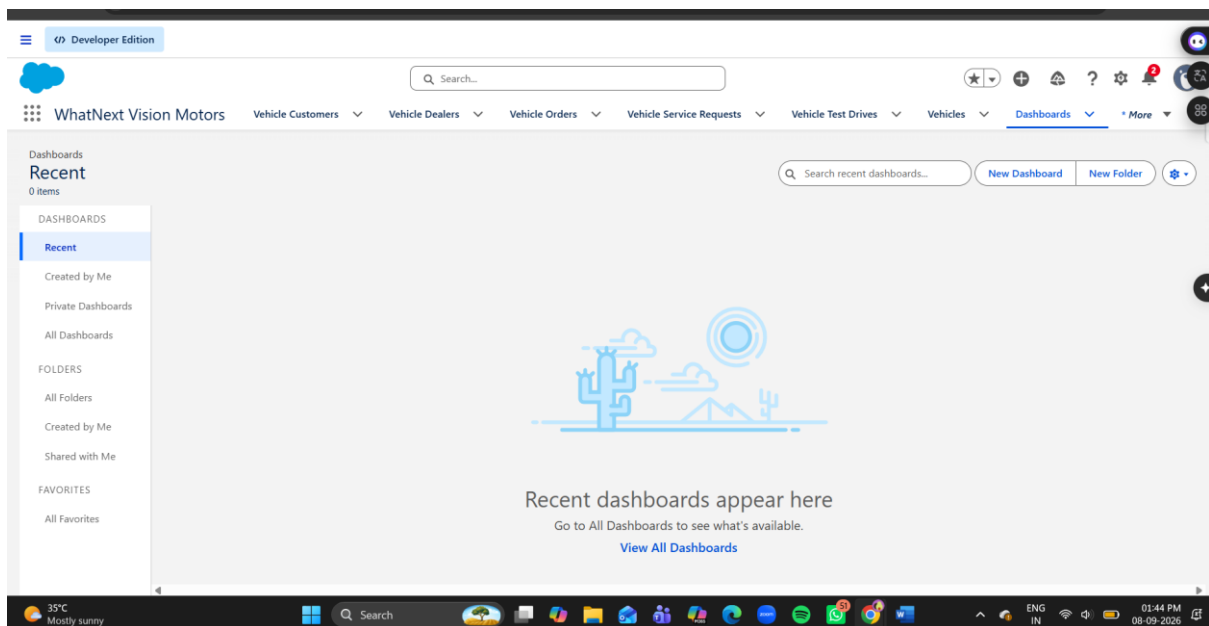
Test Case Format

Each test case contains:

- Test Case ID
- Feature
- Input
- Expected Result
- Actual Result
- Status
- Screenshot Evidence

Sample Test Cases

ID	Feature	Test Condition	Expected Result
TC-001	Vehicle	Create valid vehicle	Record created
TC-002	Stock	Enter invalid stock	Validation prevents save
TC-003	Order	Vehicle available	Order processed
TC-004	Order	Vehicle unavailable	Order prevented/handled
TC-005	Dealer	Customer location matches dealer	Dealer assigned
TC-006	Test Drive	Status = Scheduled	Reminder scheduled
TC-007	Batch	Stock = 0	Availability updated
TC-008	Report	Generate order report	Correct records displayed



PHASE 5 – DEPLOYMENT, DOCUMENTATION & MAINTENANCE

Deployment Strategy

Salesforce components can be moved between environments using an appropriate deployment mechanism such as Change Sets or Salesforce-supported development and deployment tools.

Deployment Components

- Custom Objects
- Custom Fields
- Validation Rules
- Flows

- Apex Classes
- Apex Triggers
- Permission Sets
- Lightning Pages
- Reports
- Dashboards

Deployment Validation

Before deployment, the following should be checked:

- Configuration dependencies
- Apex test coverage
- Flow activation
- Permissions
- Lookup relationships
- Required fields
- Reports
- Automation dependencies

Maintenance

Regular maintenance should include:

- Monitoring failed flows.
- Reviewing Apex Jobs.
- Monitoring scheduled jobs.
- Reviewing debug logs when necessary.
- Checking data quality.
- Reviewing user permissions.
- Updating reports and dashboards.
- Maintaining documentation.

Troubleshooting Approach

Identify Issue → Reproduce Issue → Review Logs → Identify Root Cause → Apply Fix → Retest → Document Resolution

21. PROJECT BENEFITS AND FUTURE ENHANCEMENTS

Benefits

Improved Customer Experience

Customers receive a more structured and efficient journey from vehicle selection to order processing.

Better Inventory Visibility

Vehicle stock information is centralized and can be monitored through Salesforce.

Reduced Manual Work

Flows and Apex automation reduce repetitive administrative activities.

Faster Dealer Allocation

Location-based dealer assignment reduces manual dealer selection.

Improved Communication

Automated test-drive reminders provide timely customer communication.

Better Operational Control

Order status and stock information provide employees with improved visibility.

Improved Decision Making

Reports and dashboards provide management with actionable information.

Future Enhancements

The current project can be extended with additional Salesforce capabilities.

AI-Powered Vehicle Recommendations

An AI-based recommendation system could suggest vehicles based on customer preferences, budget and previous interactions.

Chatbot Integration

A Salesforce chatbot could answer customer questions regarding vehicles, availability, dealers and test drives.

Real-Time Geographic Dealer Matching

A future version could use latitude/longitude coordinates and distance calculations to identify the physically nearest dealer rather than relying only on location matching.

Customer Portal

A customer-facing portal could allow customers to:

- Browse vehicles.
- Schedule test drives.
- Track orders.

- Submit service requests.
- View service status.

Mobile Application

Salesforce mobile capabilities could provide dealers and sales representatives with access to customer and order information while working remotely.

22. CONCLUSION

The **WhatsNext Vision Motors Salesforce CRM Project** demonstrates how Salesforce can be used to design and implement a centralized automotive CRM solution that combines data management, automation, custom development and analytics.

The project addresses key automotive business requirements including vehicle inventory management, dealer management, customer management, vehicle ordering, test-drive scheduling and service-request management.

The use of Salesforce Flow enables repetitive business processes to be automated, while Apex and triggers provide programmatic enforcement of business rules. Batch Apex and Scheduled Apex extend the system's capability to process records asynchronously and execute recurring operations.

The Lightning application provides users with a centralized interface for accessing CRM information, while reports and dashboards provide management with better visibility into operational activities.

The project also demonstrates the importance of testing, security, data quality and maintainable architecture in a Salesforce implementation.

Overall, the proposed solution provides a strong foundation for improving operational efficiency, reducing manual effort, improving customer communication and supporting future digital transformation initiatives at WhatsNext Vision Motors.